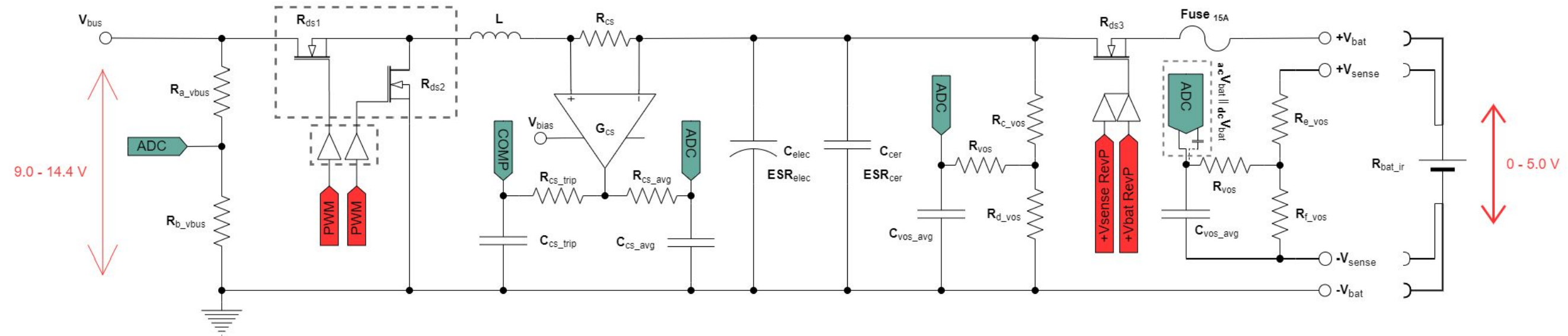




Texas Instruments and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. Texas Instruments and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. Texas Instruments and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.

1	2	3	4	5	6
A					
B					
C					
D					

MAIN SHEET
XTIDA-010086_MAIN SHEET.SchDoc

CoverSheet
XTIDA-010086_CoverSheet.SchDoc

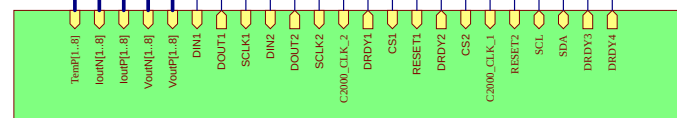
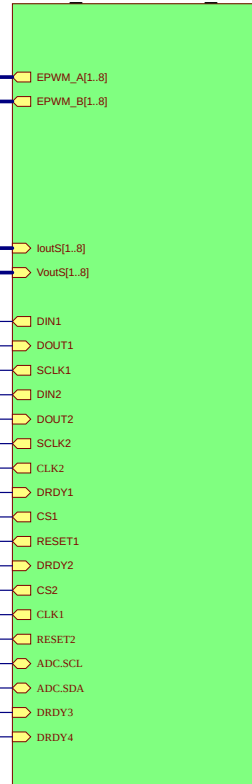
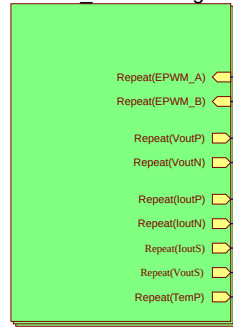
Hardware
XTIDA-010086_Hardware.SchDoc

Texas Instruments and/or its licensors do not warrant the accuracy or completeness of this specification or any information contained therein. Texas Instruments and/or its licensors do not warrant that this design will meet the specifications, will be suitable for your application or fit for any particular purpose, or will operate in an implementation. Texas Instruments and/or its licensors do not warrant that the design is production worthy. You should completely validate and test your design implementation to confirm the system functionality for your application.			Orderable:	Designed for: Public Release	Mod. Date: 27/12/2019
			TID #:	010086	Project Title: 10A Digital battery test system
			Number:	XTIDA-010086	Rev: E3
			SVN Rev:	d5b8d99659a9160a49f82a35d110047aac69081	Sheet 2 of 8
			Drawn By:	Jeff Xie	File: XTIDA-010086_TOP LEVEL MAIN SHEET.SchDoc
			Engineer:	Jeff Xie / Shaury Anand / Luciano	Size: B

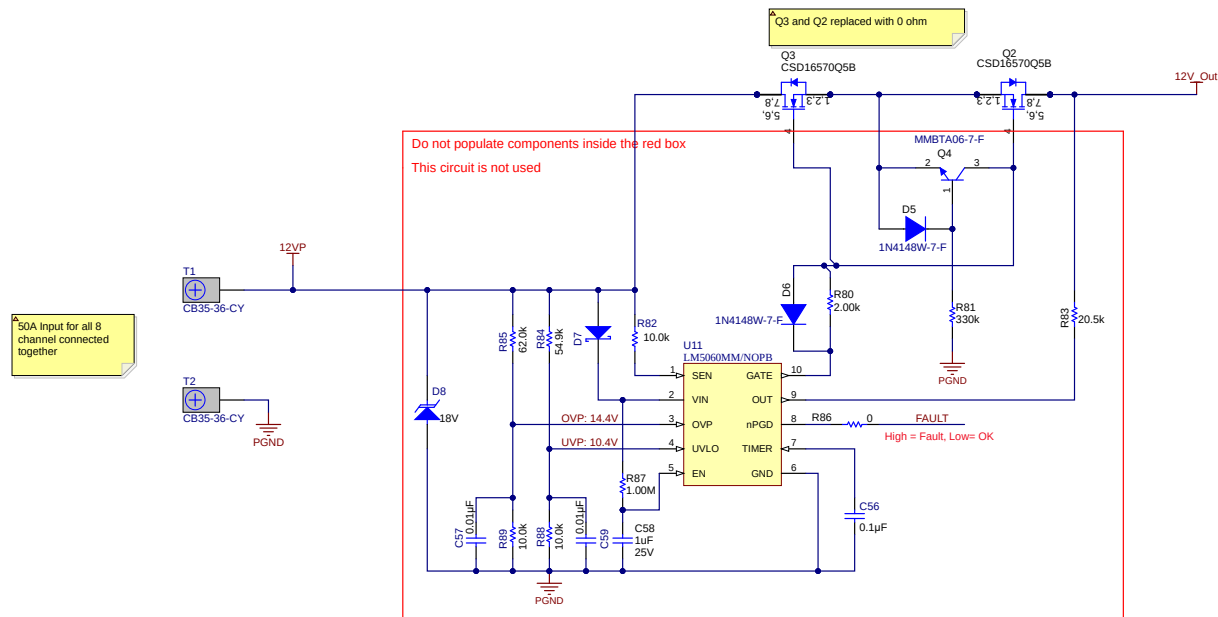
 **TEXAS
INSTRUMENTS**

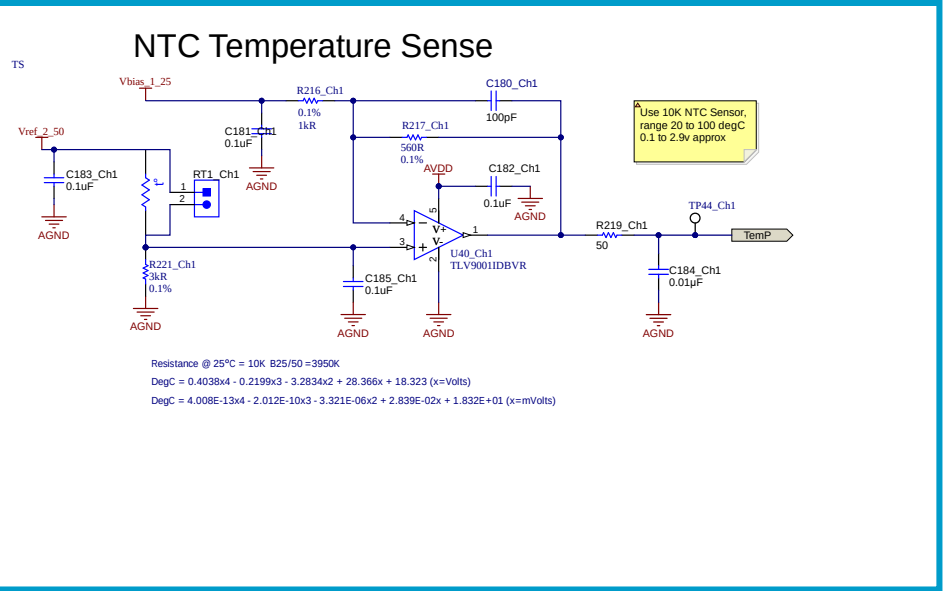
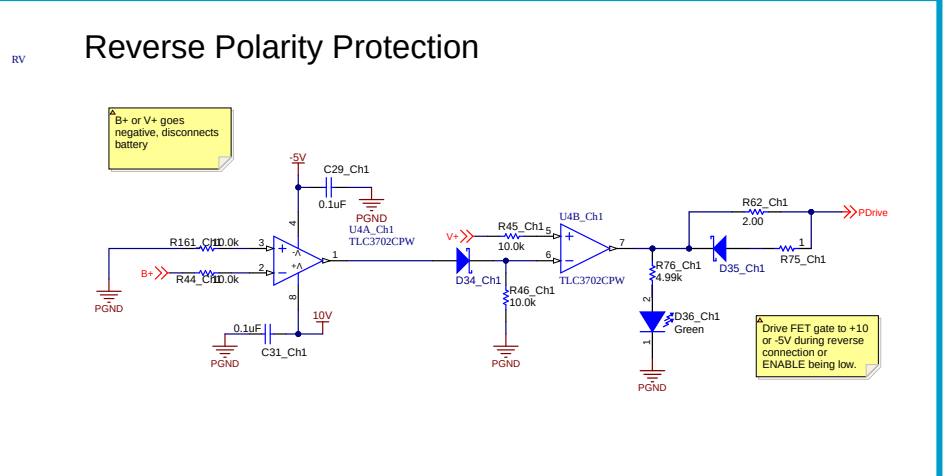
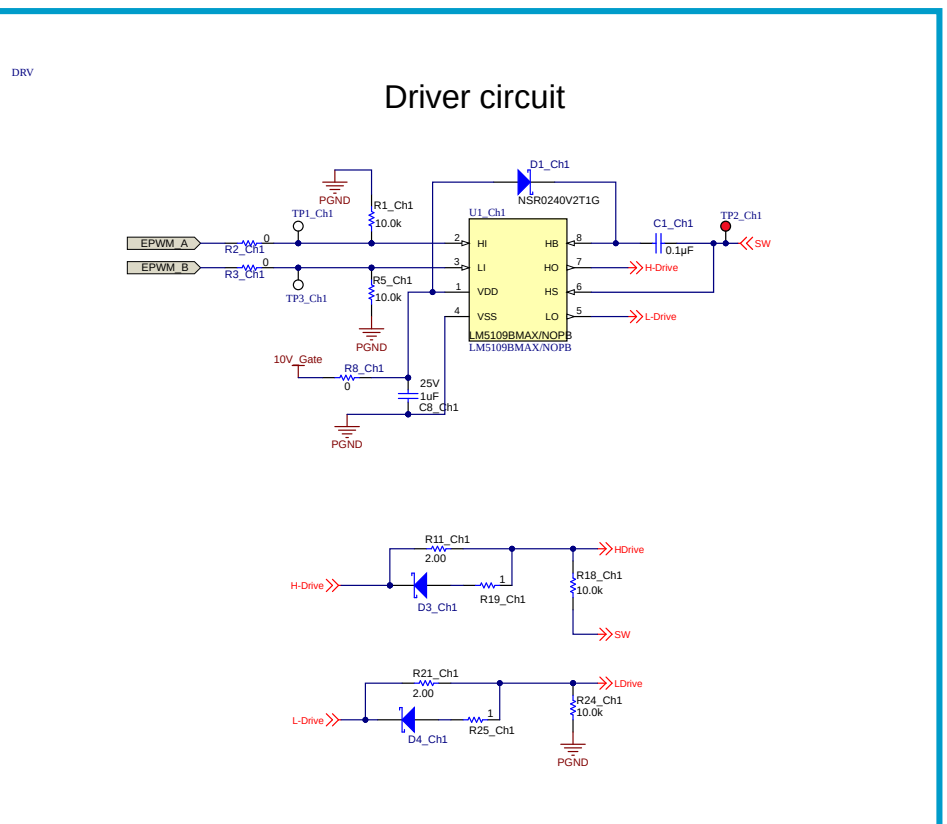
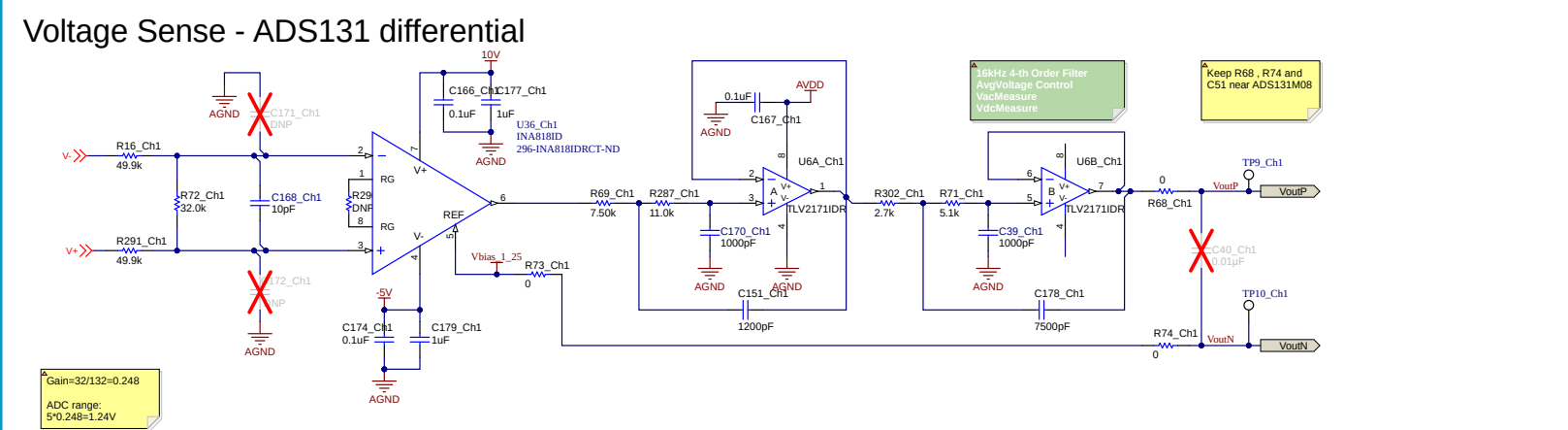
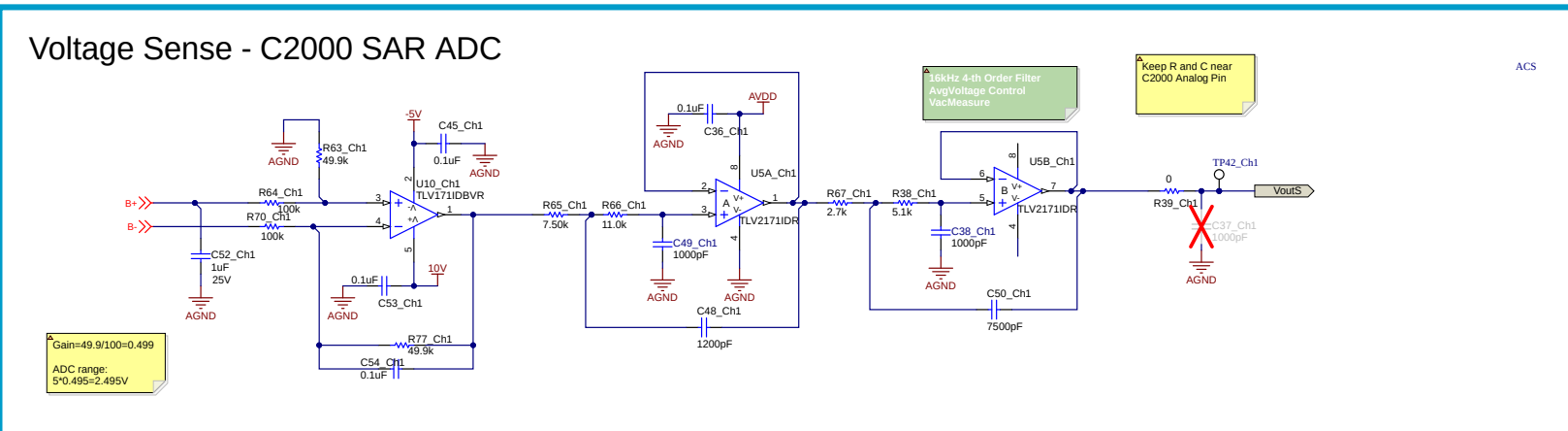
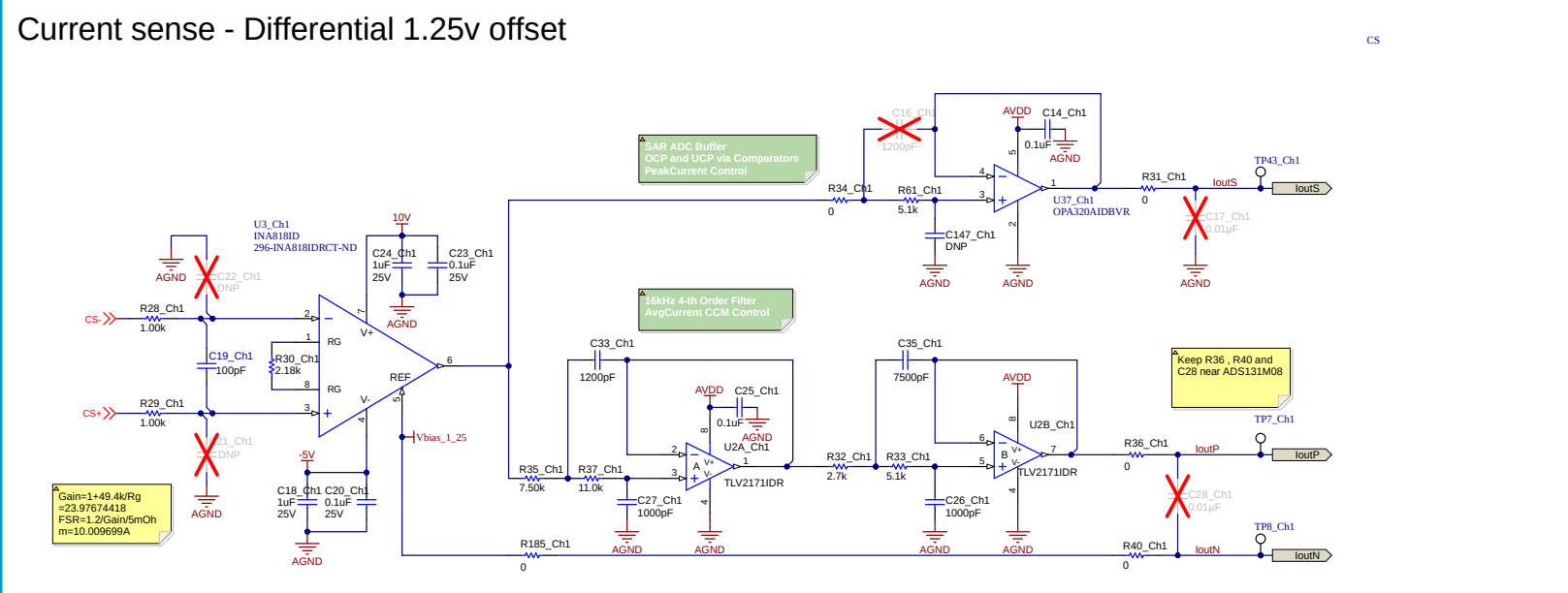
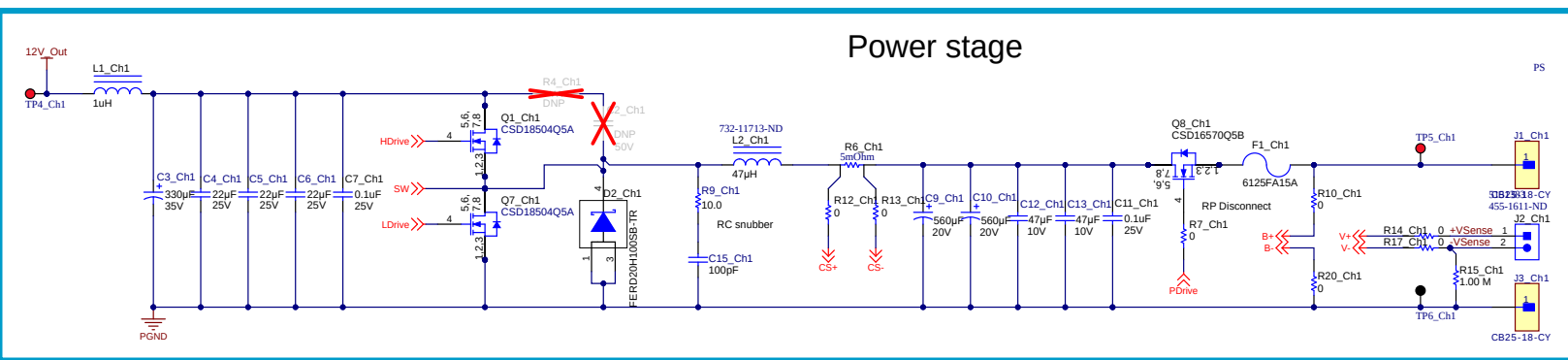
http://www.ti.com

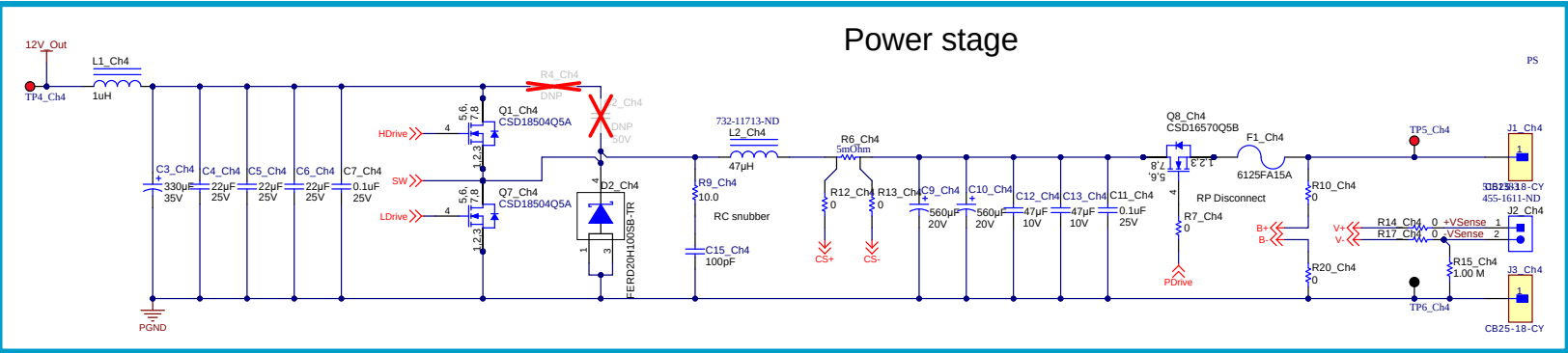
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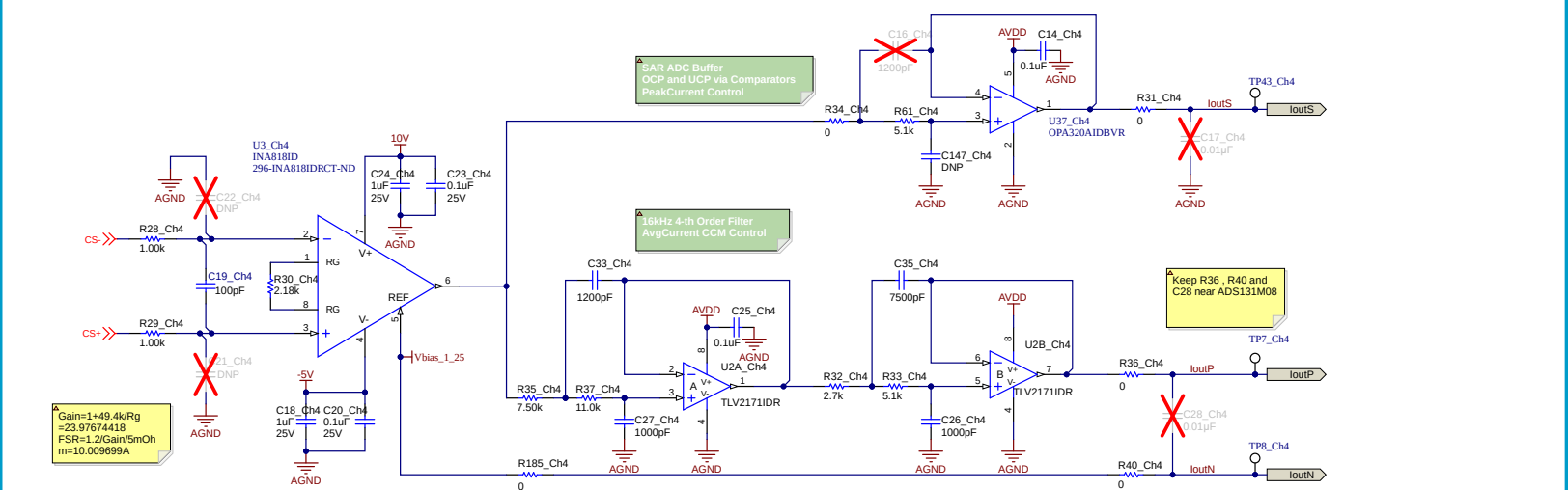
50A Input for all 8 channel connected together



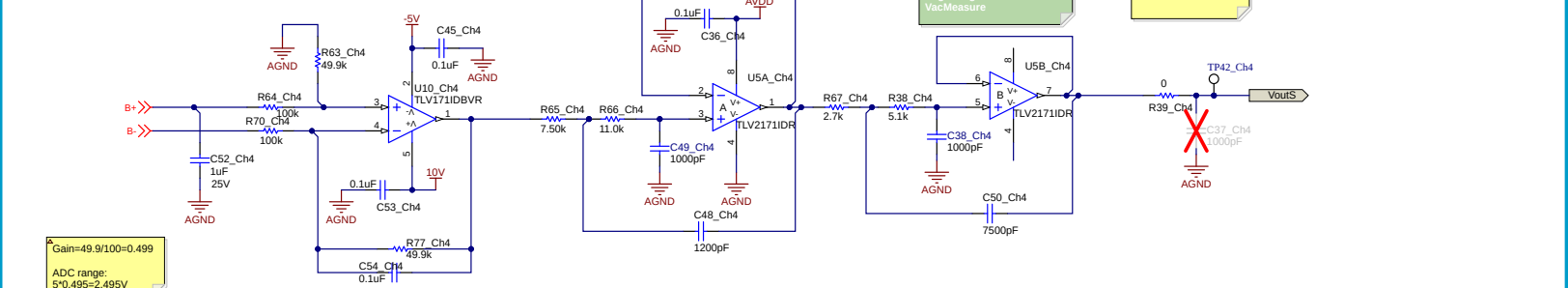




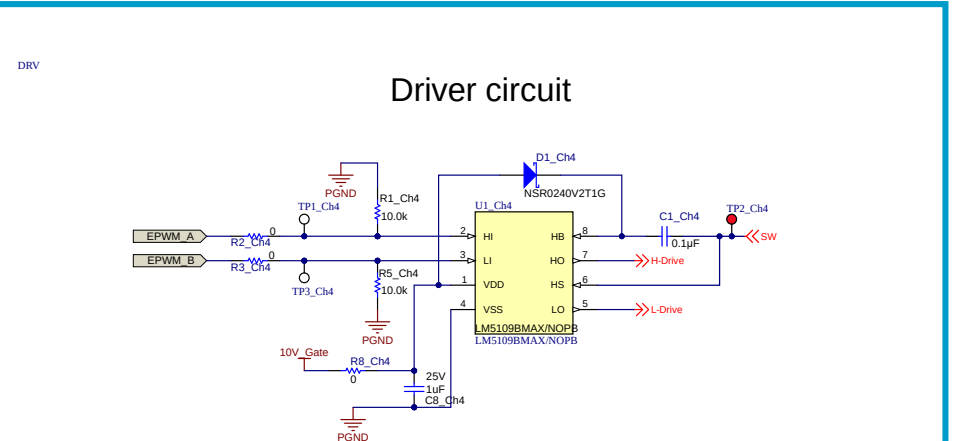
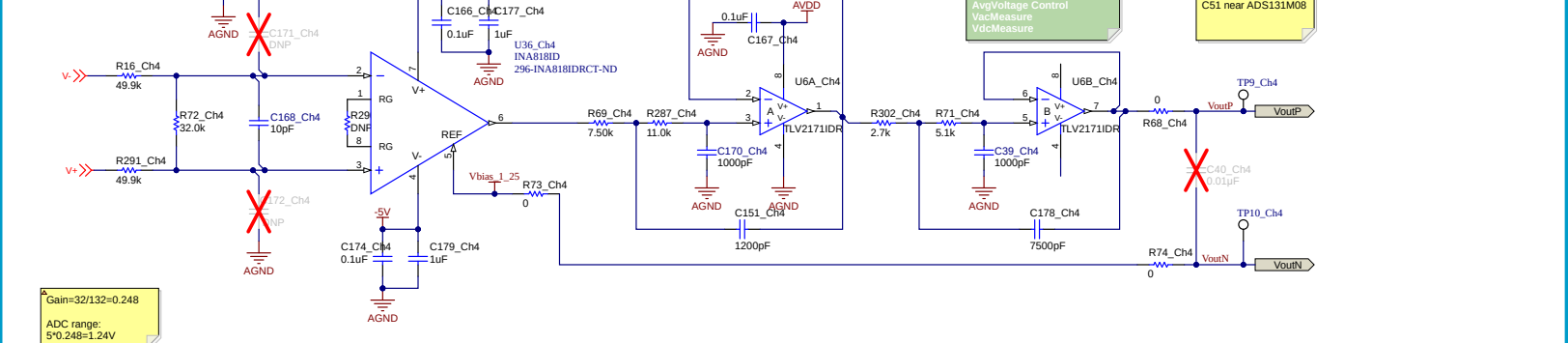
Current sense - Differential 1.25v offset



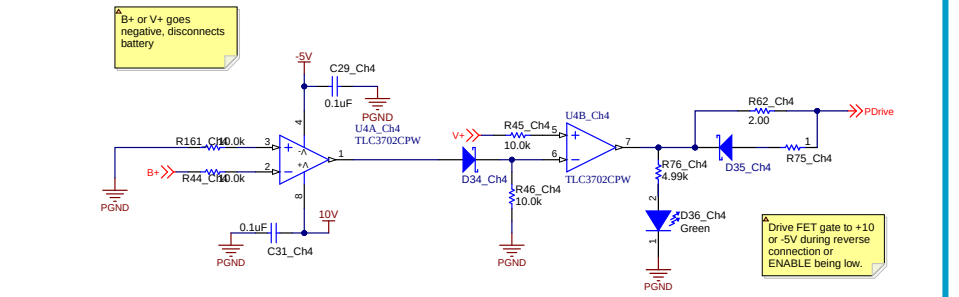
Voltage Sense - C2000 SAR ADC



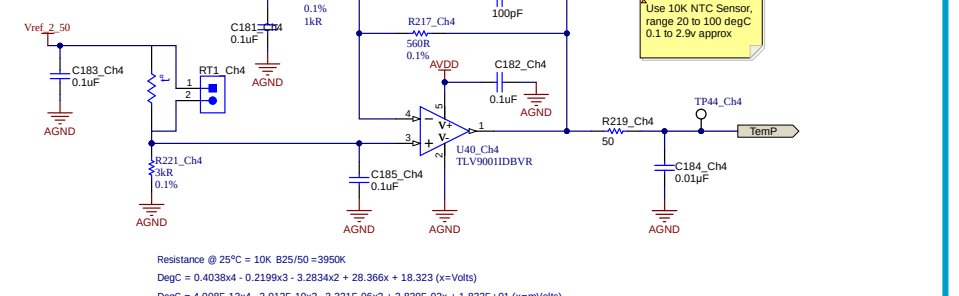
Voltage Sense - ADS131 differential



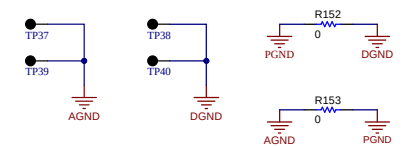
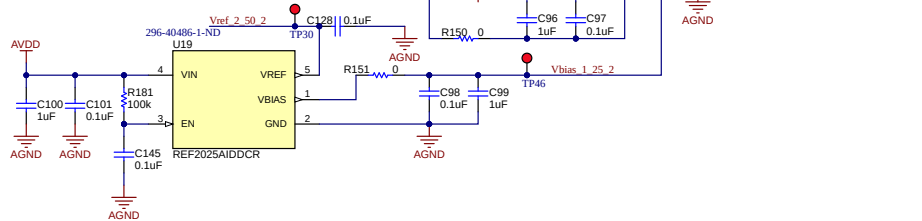
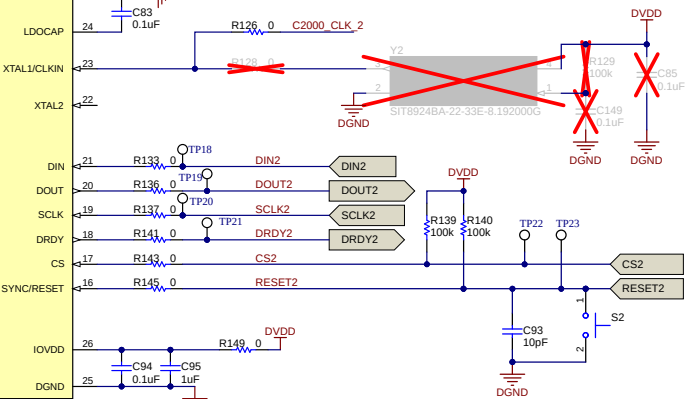
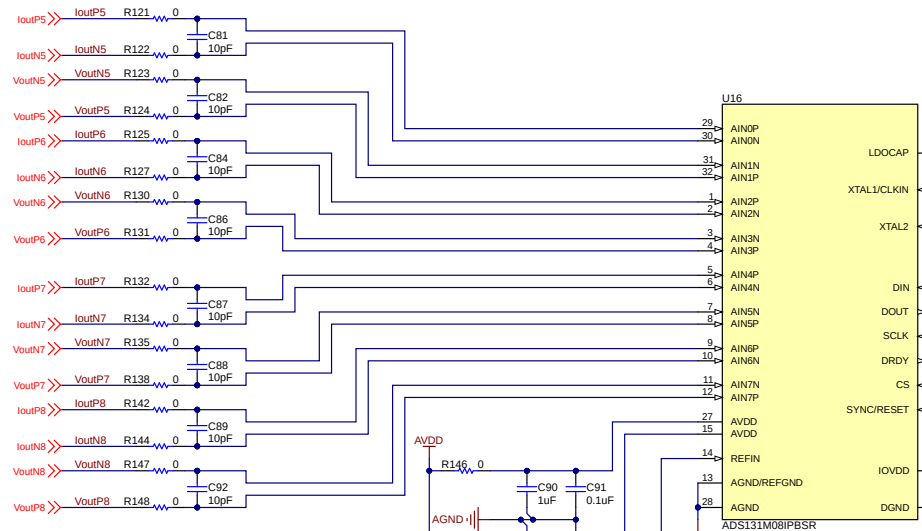
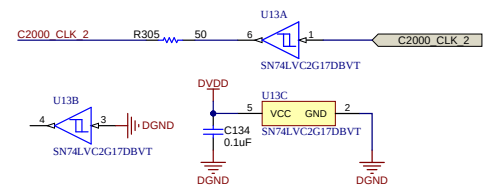
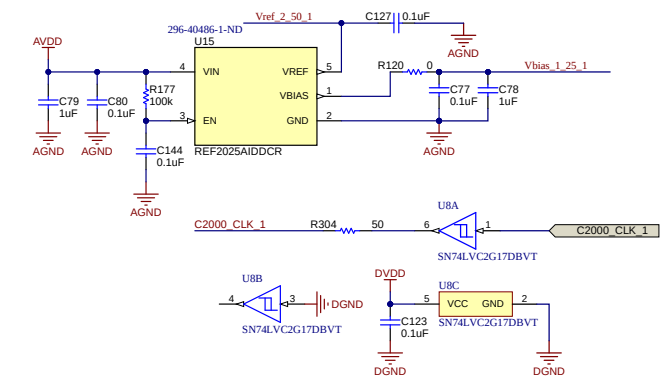
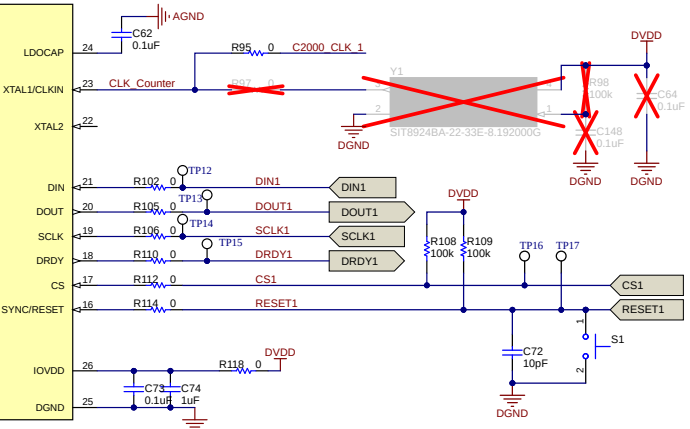
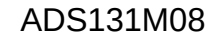
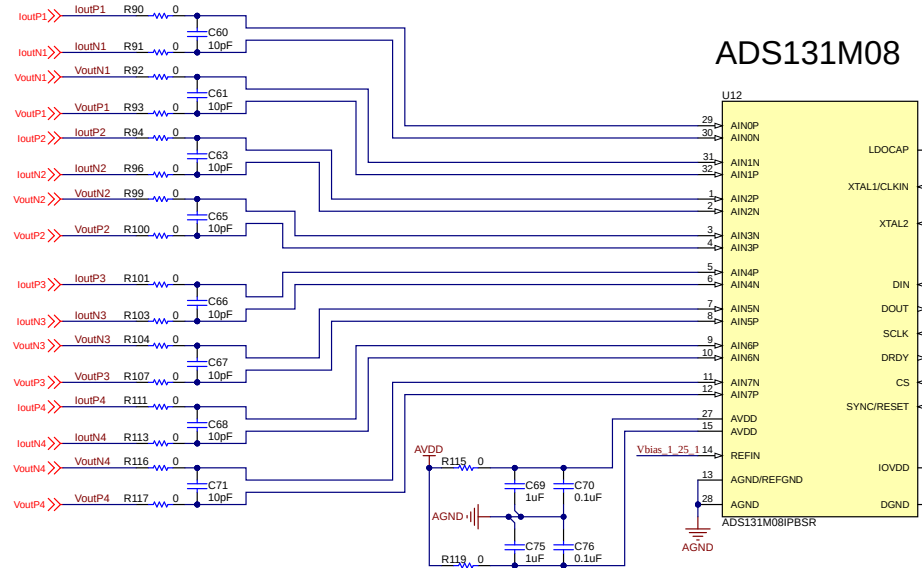
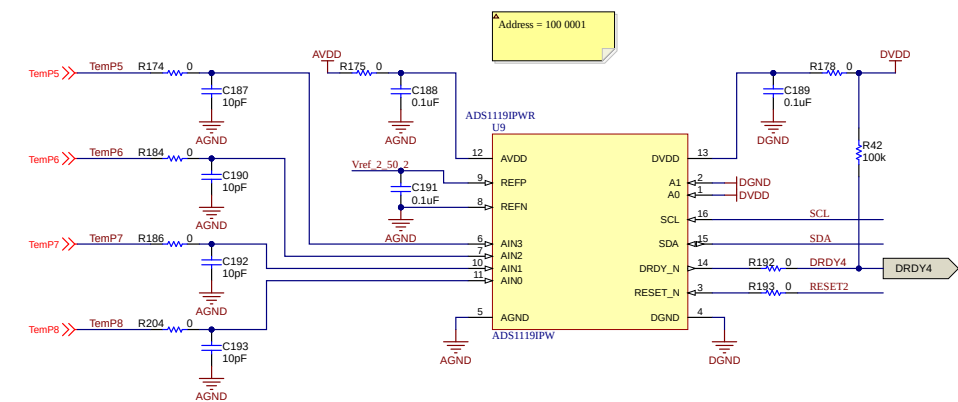
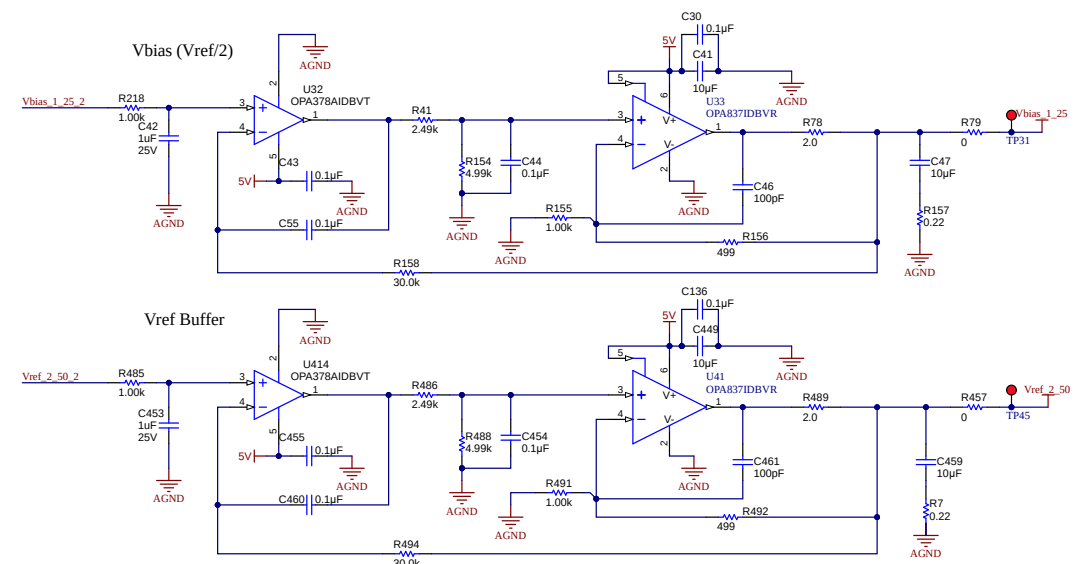
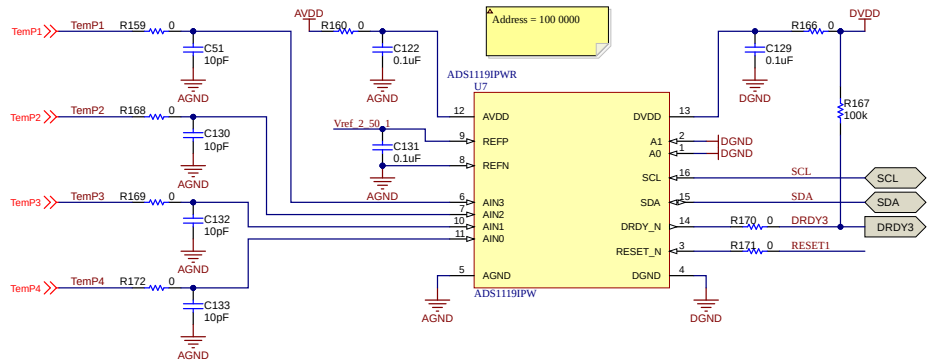
Reverse Polarity Protection



NTC Temperature Sense



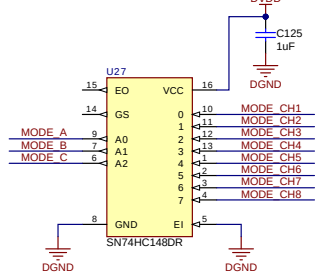
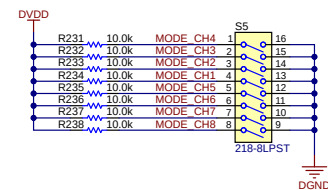
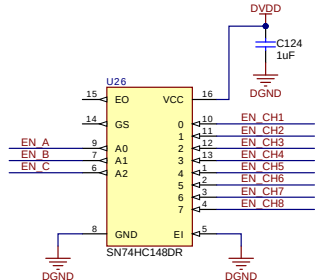
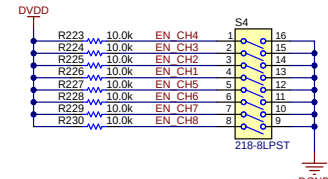
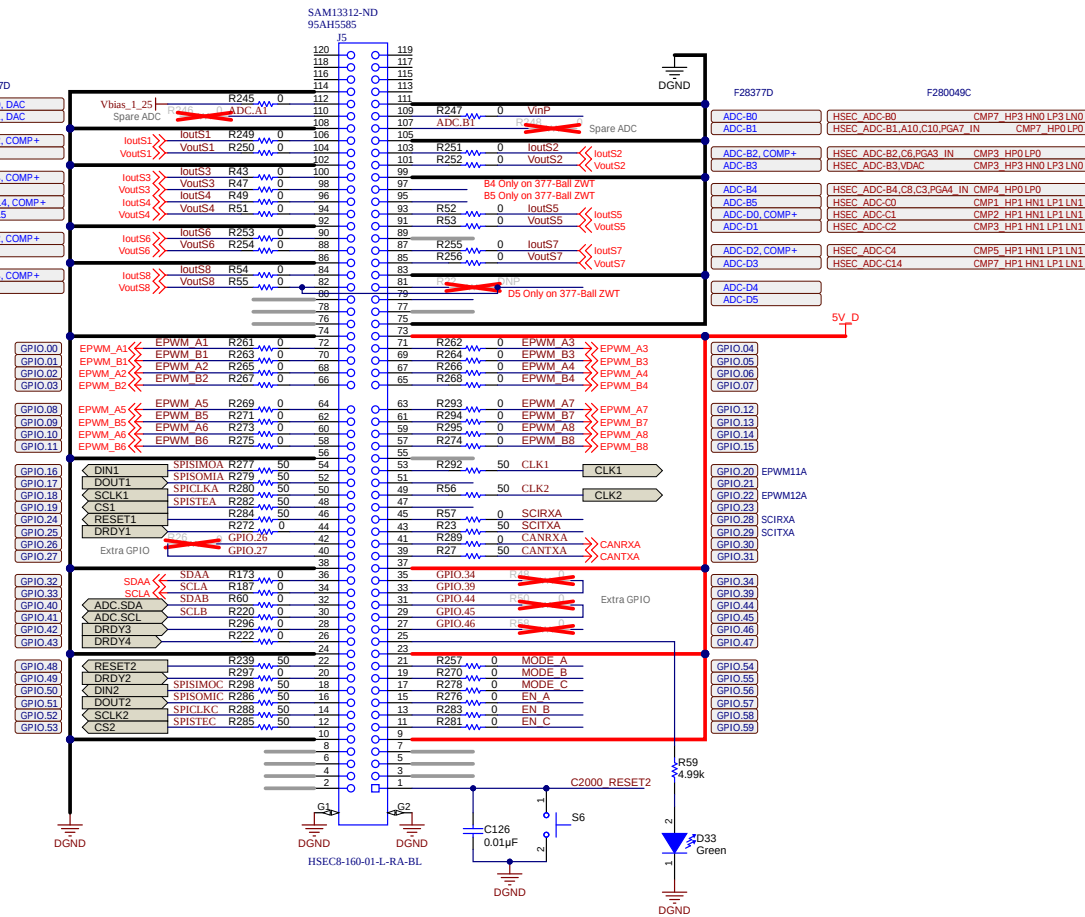
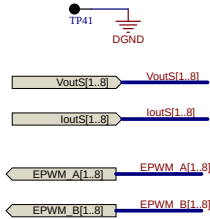
ADS1119



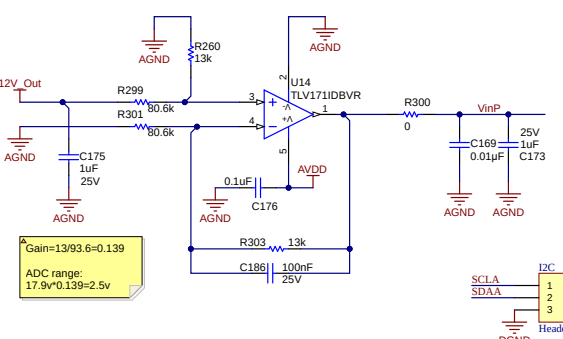
IoutS[1,3] -> ADC-A
IoutS[2,4] -> ADC-B
IoutS[5,7] -> ADC-D
IoutS[6,8] -> ADC-C
Vbias_1_25 -> ADC A0,B12,C12,D12

CONTROL CARD PIN MAPPING_28379 (RA Connector)

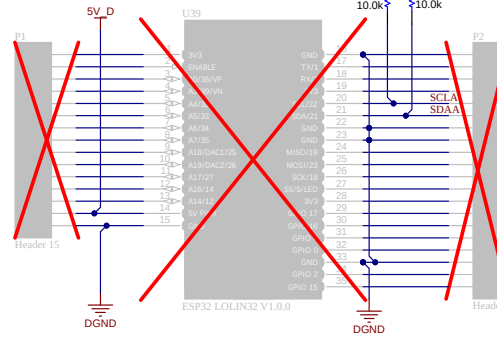
Map ADCIN14/15 -> ADC-A14/15



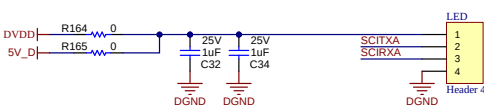
Input voltage monitoring



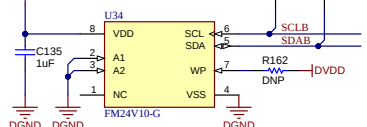
ESP32 WiFi External Iface



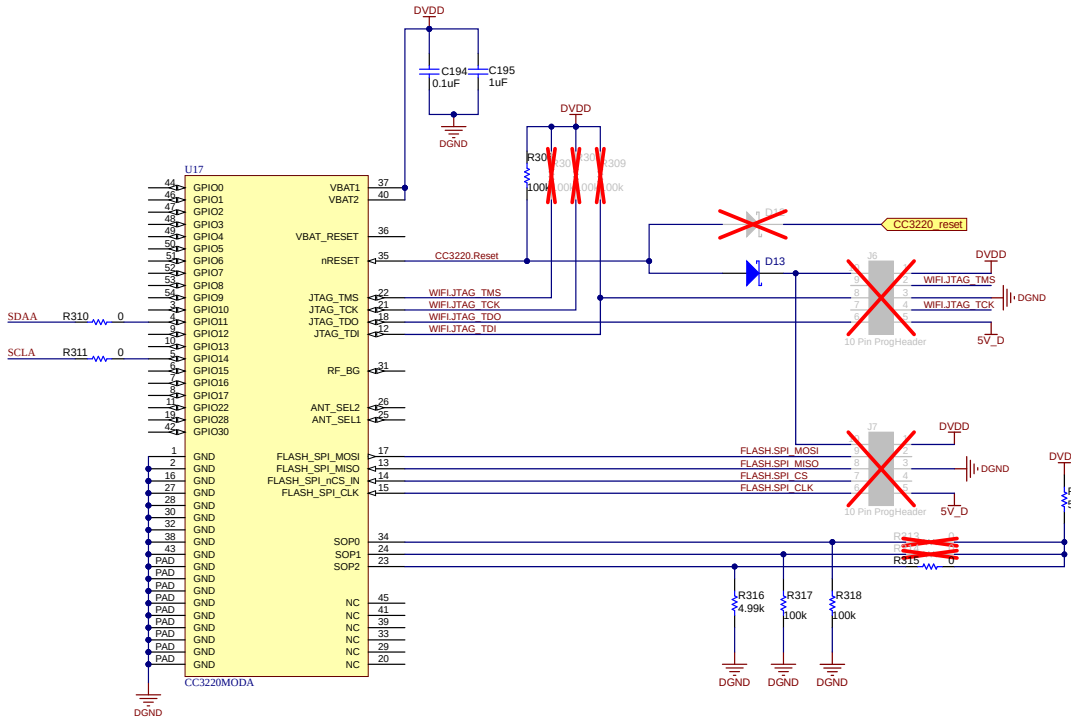
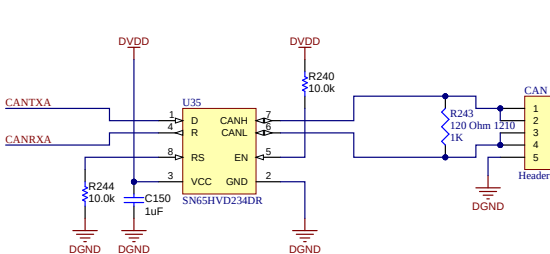
External LED Strip, 3.3 or 5v



Board Calibration Data



External Canbus





PCB
LOGO
FCC disclaimer

PCB
LOGO
WEEE logo



PCB
LOGO
Pb-Free Symbol

PCB
LOGO
Logo7

LBL1
THT-14-423-10
Size: 0.65" x 0.20 "

ZZ1
This Assembly Note is for PCB labels only

ZZ2
These assemblies are ESD sensitive, ESD precautions shall be observed.

ZZ3
These assemblies must be clean and free from flux and all contaminants. Use of no clean flux is not acceptable.

ZZ4
These assemblies must comply with workmanship standards IPC-A-610 Class 2, unless otherwise specified.

Variant/Label Table	
Variant	Label Text
001	ChangeMe!
002	Ti TIDA RLS Version
003	BadBiki Version